Exercise 13

Solution 1:

Steps of construction:

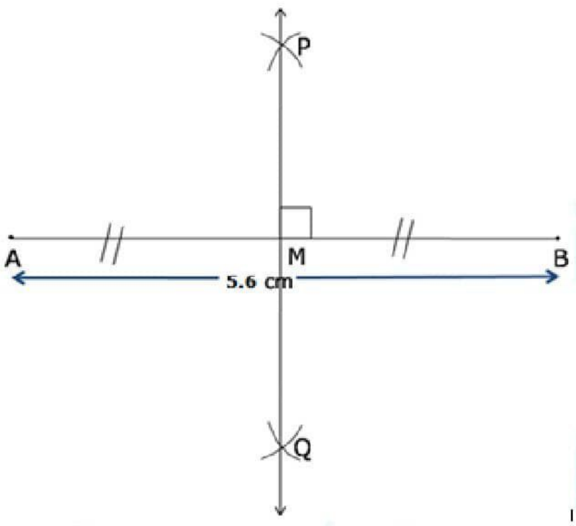
(i) Draw line segment AB = 5.6 cm

(ii) With A as centre and radius more than half of AB, draw two arcs, one on each side of AB.

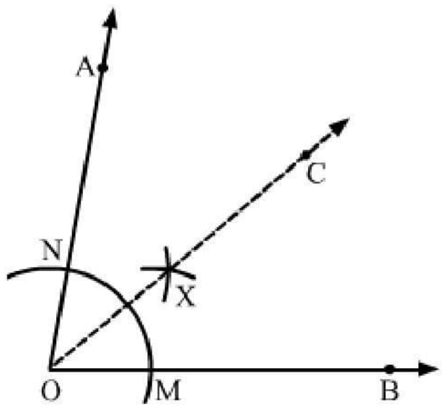
(iii) With B as centre and the same radius as in step 2, draw arcs cutting the arcs drawn in the previous step at P and Q respectively.

(iv) Join PQ to intersect AB at M.

Thus, PQ is the required perpendicular bisector of AB.



Solution 2:



Steps of construction

1. Draw $\angle AOB = 80°$ using protractor.

2. With O as centre and a convenient radius, draw an arc cutting AO at N and OB at M.

3. With N as centre and a convenient radius, draw an arc.

4. Similarly, with M as centre and same radius, cut the previous drawn arc and name it as point C.

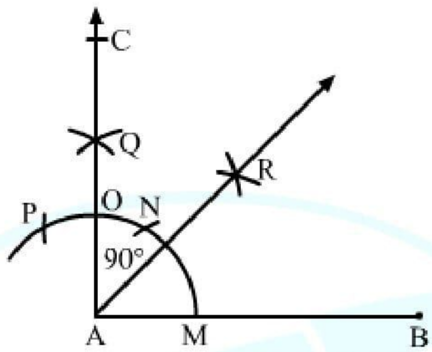
5. Join OC.

OC is the required angle bisector.

On measuring we get

$\angle AOC = \angle BOC = 40°$

Solution 3:



Steps of construction:

1. Draw a line segment AB.

2. With A as the centre and a small radius, draw an arc cutting AB at M.

3. With M as the centre and the same radius as above, draw an arc cutting the previously drawn arc at N.

4. With N as the centre and the same radius as above, draw an arc cutting the previously drawn arc at P.

5. Again, with N as the centre and a radius more than half of PN, draw an arc.

6. With P as the centre and the same radius as above, draw an arc cutting the previously drawn arc at Q.

7. Join AQ cutting the arc at O and produced it to C.

8. With O as the centre and a radius more than half of OM, draw an arc.

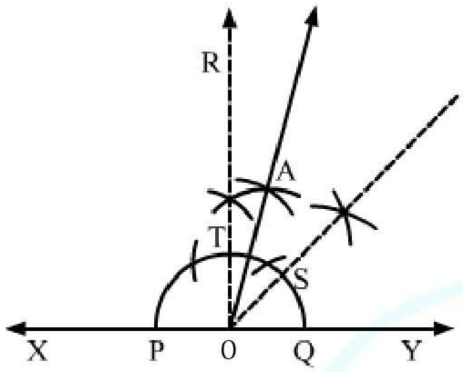
9. With M as the centre and the same radius as above, draw another arc cutting the previously drawn arc at point R.

10. Join AR.

Thus, AR bisects ∠BAC.

Solution 4:

(i) 75°

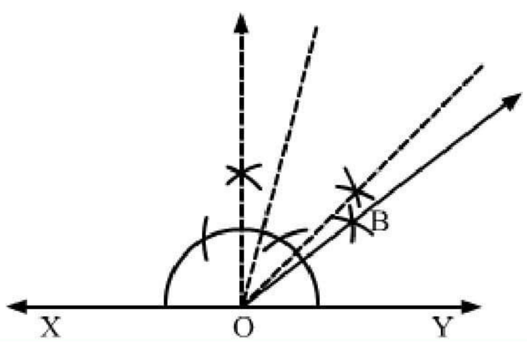


Steps of construction

1. Draw a line XY.
2. Take a point O on XY.
3. With O as centre, draw a semi circle, cutting XY at P and Q.
4. Construct ∠YOR = 90°.
5. Draw the bisector of ∠YOR = 90°° cutting the semi circle at point S.
6. With S and T as centres draw two arcs intersecting at point A.

∠AOY = 75°.

(ii) 37.5°



Steps of construction

1. Draw a line XY.

2. Take a point O on XY.

3. With O as centre, draw a semi circle, cutting XY at P and Q.

4. Construct ∠YOR = 90°.

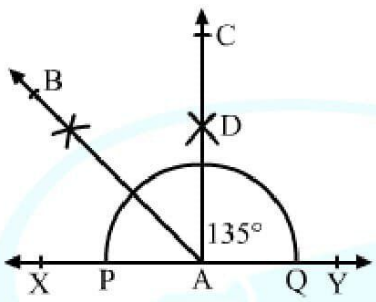
5. Draw the bisector of ∠YOR = 90°° cutting the semi circle at point S.

6. With S and T as centres draw two arcs intersecting at point A.

7. Draw the angle bisector of ∠AOY.

8. ∠BOY is the required angle of 37.5°.

(iii) 135°



Steps of construction:

1. Draw a line XY.

2. Take a point A on XY.

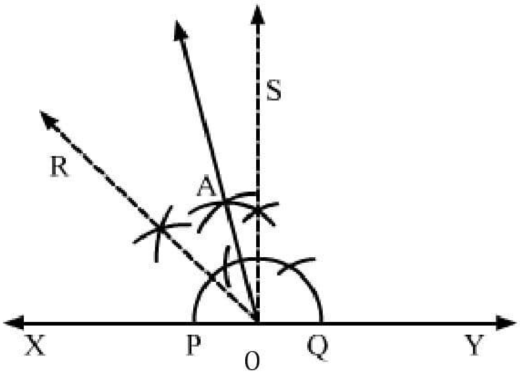
3. With A as centre, draw a semi circle, cutting XY at P and Q.

4. Construct ∠YAC = 90°°.

5. Draw AB, bisector of ∠XAC.

Thus, ∠YAB = 135°°

(iv) 105°



Steps of construction

1. Draw a line XY.

2. Take a point O on XY.

3. With O as centre, draw a semi circle, cutting XY at P and Q.

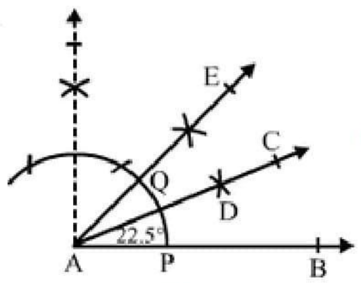
4. Construct ∠YOS = 90°°.

5. Draw RO, bisector of ∠XOS.

6. Draw AO, bisector of ∠ROS.

∠AOY = 105° is the required angle.

(v) 22.5°



Steps of construction:

1. Draw a ray AB.

2. Draw an angle ∠∠BAE = 45°°.

3. With A as the centre and a small radius, draw an arc cutting AB at P and AE at Q.

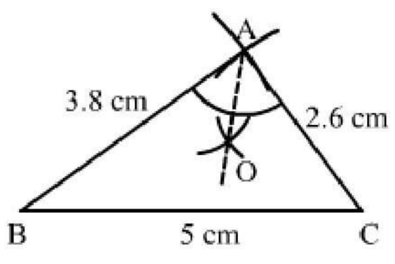
4. With P as the centre and a radius more than half of PQ, draw an arc.

5. With Q as the centre and the same radius as above, draw another arc cutting the previously drawn arc at D.

6. Join AD.

Thus, ∠∠BAC is the required angle of measure 22.5o.

Solution 5:



Steps of construction:

1. Draw a line segment BC = 5 cm.

2. With B as the centre and a radius equal to 3.8 cm, draw an arc.

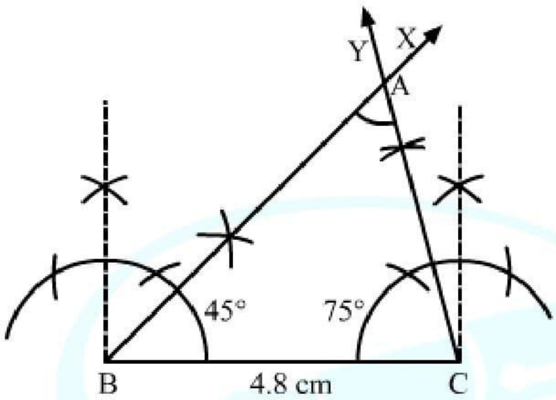
3. With C as the centre and a radius equal to 2.6 cm, draw an arc cutting the previously drawn arc at A.

4. Join AB and AC.

Thus, ABC is the required triangle.

Take the largest angle and draw the angle bisector.

Solution 6:



Steps of construction:

1. Draw a line segment BC = 4.8 cm.

2. Construct ∠CBX = 45°.

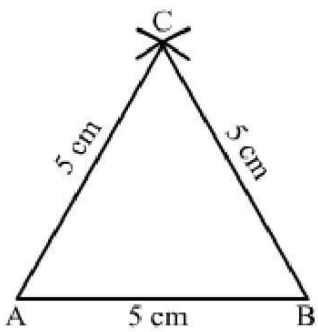
3. Construct ∠BCY = 75°.

4. The ray BX and CY intersect at A.

Thus, △ABC is the required triangle.

When we measure ∠A, we get ∠A = 60°.

Solution 7:



Steps of construction:

1. Draw a line segment AB = 5 cm.

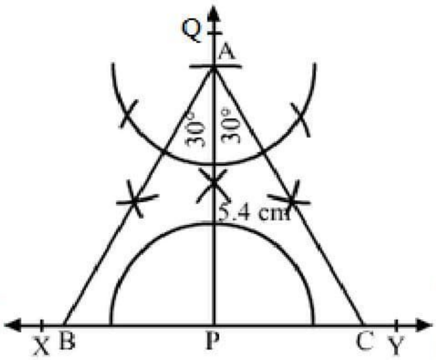
2. With A as the centre and a radius equal to AB, draw an arc.

3. With B as the centre and the same radius as above, draw another arc cutting the previously drawn arc at C.

4. Join AC and BC.

Thus, △△ABC is the required triangle.

Solution 8:



Steps of construction:

1. Draw a line XY.

2. Mark any point P.

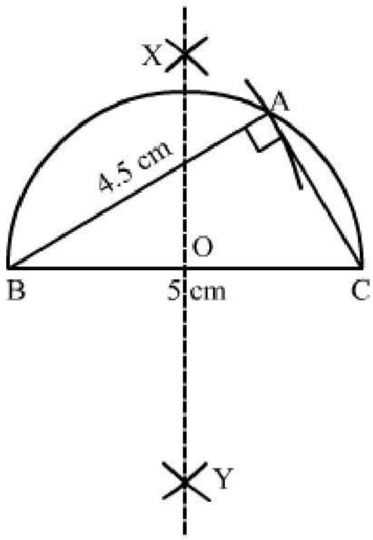
3. From P draw PQ ⊥ XY.

4. From P, set off PA = 5.4 cm, cutting PQ at A.

5. Construct ∠PAB = 30° and ∠PAC = 30°, meeting XY at B and C, respectively.

Thus, △ABC is the required triangle. Measure of each side is 6 cm.

Solution 9:



Steps of construction:

1. Draw a line segment BC = 5 cm.

2. Find the midpoint O of BC.

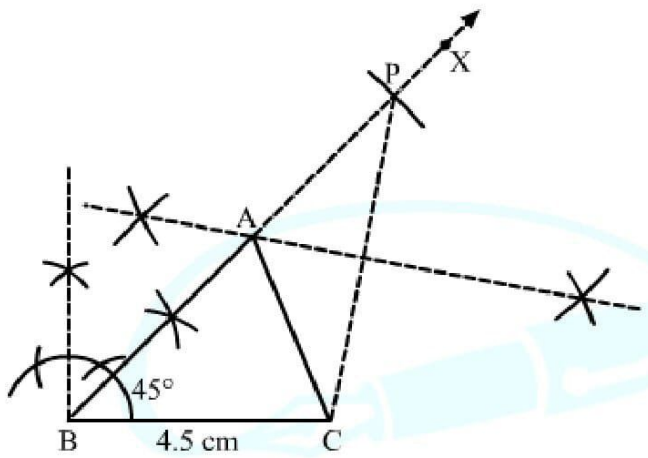
3. With O as the centre and radius OB, draw a semicircle on BC.

4. With B as the centre and radius equal to 4.5 cm, draw an arc cutting the semicircle at A

4. Join AB and AC.

Thus, ABC is the required triangle.

Solution 10:



Steps of construction:

1. Draw a line segment BC = 4.5 cm.

2. Construct ∠CBX = 45°.

3. Set off BP = 8 cm.

4. Join PC.

5. Draw the right bisector of PC, meeting BP at A.

6. Join AC.

Thus, △ABC is the required triangle.

Justification:

In △APC,

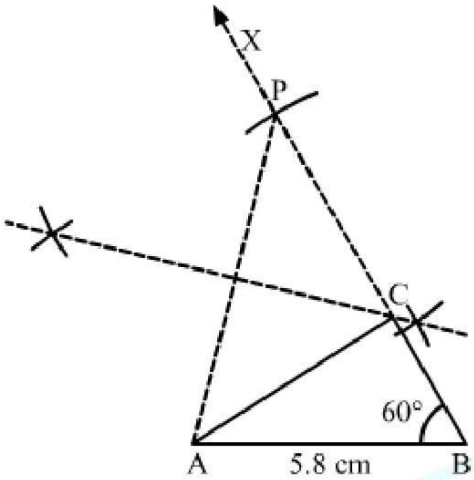
∠ACP = ∠APC	[By construction]
⇒ AC = AP	[Sides opposite to equal angles are equal]

Now

AB = BP – AP = BP – AC

⇒ AB + AC = BP

Solution 11:



Steps of construction:

1. Draw a line segment AB = 5.8 cm.
2. Construct ∠ABX = 60°.
3. Set off BP = 8.4 cm.
4. Join PA.
5. Draw the right bisector of PA, meeting BP at C.
6. Join AC.

Thus, △ABC is the required triangle.

Justification:

In △APC,

∠CAP = ∠CPA [By construction]

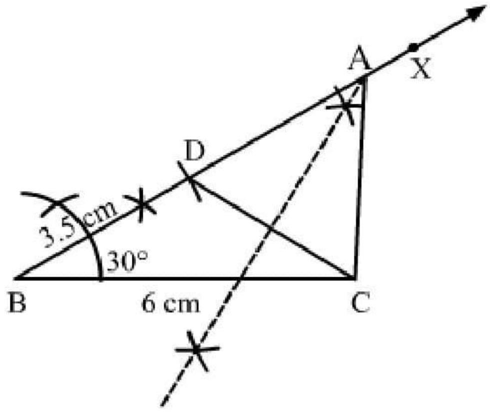
⇒ CP = AC [Sides opposite to equal angles are equal]

Now

BC = PB – PC = PB – AC

⇒ BC + AC = PB

Solution 12:



Steps of construction:

1. Draw a line segment BC = 6 cm.

2. Construct ∠CBX = 30°.

3. Set off BD = 3.5 cm.

4. Join DC.

5. Draw the right bisector of DC, meeting BD produced at A.

6. Join AC.

Thus, △ABC is the required triangle.

Justification:

Point A lies on the perpendicular bisector of DC.

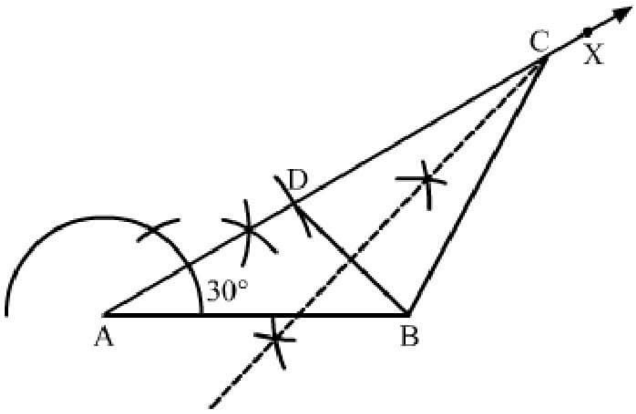
⇒ AD = AC

Now

BD = AB − AD

= AB − AC

Solution 13:



Steps of construction:

1. Draw a line segment AB = 5 cm.

2. Construct ∠BAX = 30°.

3. Set off AD = 2.5 cm.

4. Join DB.

5. Draw the right bisector of DB, meeting DB produced at C.

6. Join CB.

Thus, △ABC is the required triangle.

Justification:

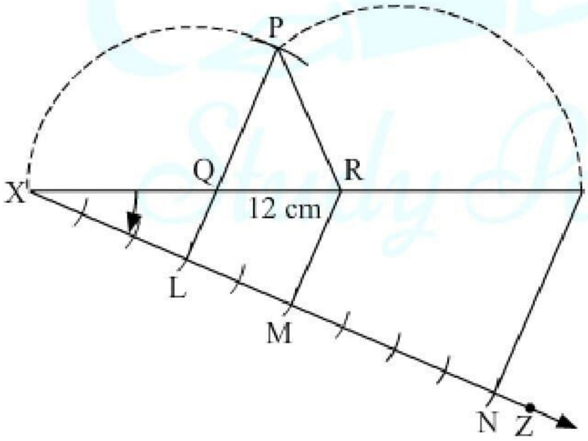
Point C lies on the perpendicular bisector of DB.

⇒ CD = BC

Now

AD = AC − DC = AC – BC

Solution 14:



1. Draw a line segment XY = 12 cm.

2. In the downward direction, construct an acute angle with XY at X.

3. From X, set off (3 + 2 + 4) = 9 arcs of equal distances along XZ.

4. Mark points L, M and N such that XL = 3 units, LM = 2 units and MN = 4 units.

5. Join NY.

6. Through L and M, draw LQ ||| NY and MR ||| NY cutting XY at Q and R, respectively.

7. With Q as the centre and radius QX, draw an arc.

8. With R as the centre and radius RY, draw an arc, cutting the previously drawn arc at

P.

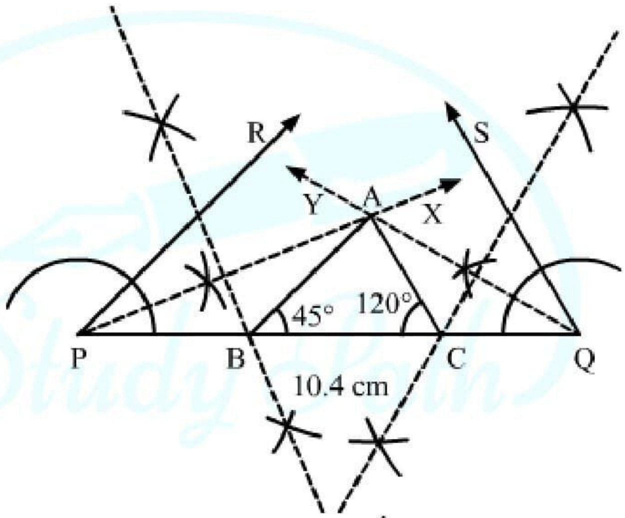
9. Join PQ and PR.

Thus, △PQR is the required triangle.

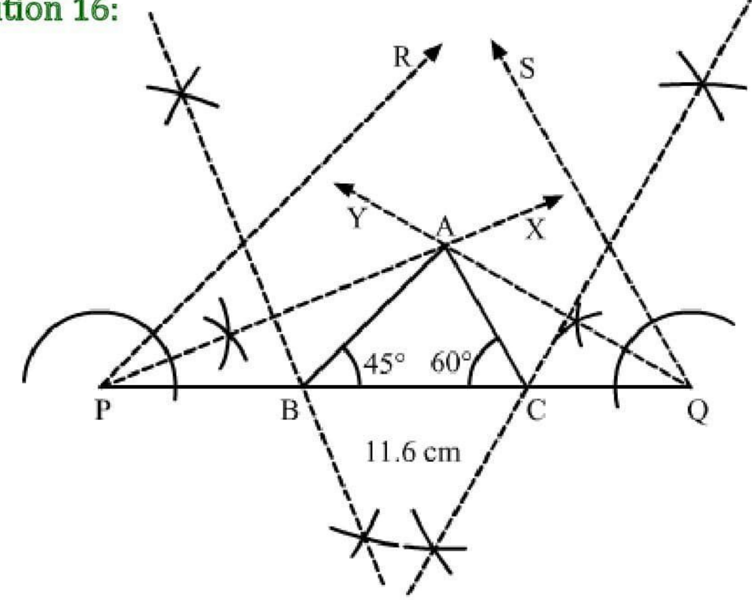
Solution 15: Steps of construction:

1. Draw a line segment PQ = 10.4 cm.
2. Construct an angle of 45° and bisect it to get ∠QPX.
3. Construct an angle of 120° and bisect it to get ∠PQY.
4. The ray XP and YQ intersect at A.
5. Draw the right bisectors of AP and AQ, cutting PQ at B and C, respectively.
6. Join AB and AC.

Thus, △ABC is the required triangle.



Solution 16:



Steps of construction:

1. Draw a line segment PQ = 11.6 cm.
2. Construct an angle of 45° and bisect it to get ∠QPX.
3. Construct an angle of 60° and bisect it to get ∠PQY.
4. The ray XP and YQ intersect at A.
5. Draw the right bisectors of AP and AQ, cutting PQ at B and C, respectively.
6. Join AB and AC.

Thus, △ABC is the required triangle.

Solution 17:

(i) AB = 6 cm, ∠A = 40° and (BC + AC) = 5.8 cm.

Here BC + AC is not greater than AB so, this triangle is not possible.

(ii) AB = 7 cm, ∠A = 50° and (BC – AC) = 8 cm.

Here BC – AC is not less than AB so this triangle is not possible.

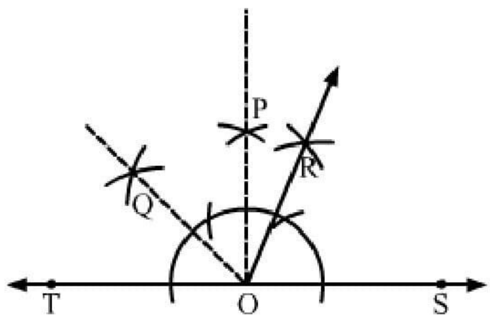
(iii) BC = 5 cm, ∠B = 80°, ∠C = 50° and ∠A = 60°.

Sum of the angles of the given triangle is not equal to 180°° so, given triangle is not possible.

(iv) AB = 4 cm, BC = 3 cm and AC = 7 cm.

Sum of two sides of this triangle not greater than third side so given triangle not possible.

Solution 18:



Steps of Construction:

1. Draw a line TS.

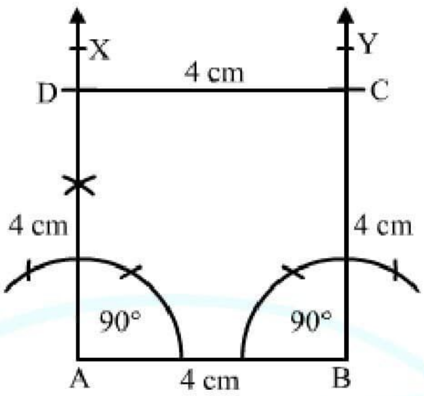
2. Construct ∠POS = 90°.

3. Draw OQ, bisector of ∠POT. Thus, ∠QOS = 135° is obtained.

4. Draw the bisector of ∠QOS.

5. ∠ROS thus obtained is equal to 67.5°.

Solution 19:



Steps of construction:

1. Draw a line segment AB = 4 cm.

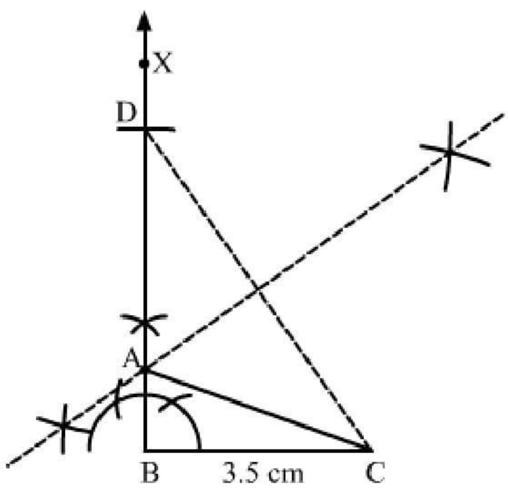
2. Construct ∠BAX = 90° and ∠ABY = 90°.

3. Set off AD = 4 cm and BC = 4 cm.

4. Join DC.

Thus, □ABCD is the required square.

Solution 20:



Steps of construction:

1. Draw a line segment BC = 3.5 cm

2. Construct ∠CBX = 90°.

3. With B as centre and 5.5 cm radius cut an arc on BX and name it as D.

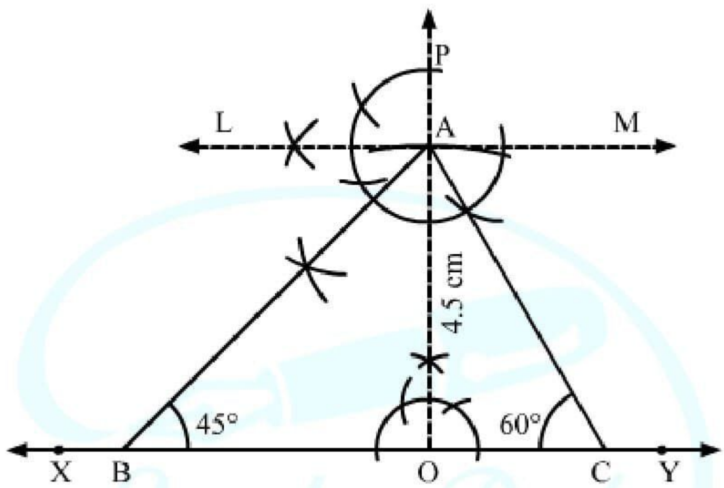
4. Join CD.

5. Construct the perpendicular bisector of CD intersecting BD at point A.

6. Join AC.

△ABC is the required triangle.

Solution 21:



Steps of construction:

1. Draw a line segment XY.

2. Take a point O on XY and draw PO ⊥⊥ XY.

3. Along PO, set off OA = 4.5 cm.

4. Draw a line LM ∥ XY.

5. Draw ∠LAB = 45° and ∠MAC = 60°, meeting XY at B and C, respectively.

Thus, ABC is the required triangle.